환경손상

경북의대 응급의학교실

안 재 윤

목차

I. Temperature regulation

II. Cold-related illness

III. Heat-related illness

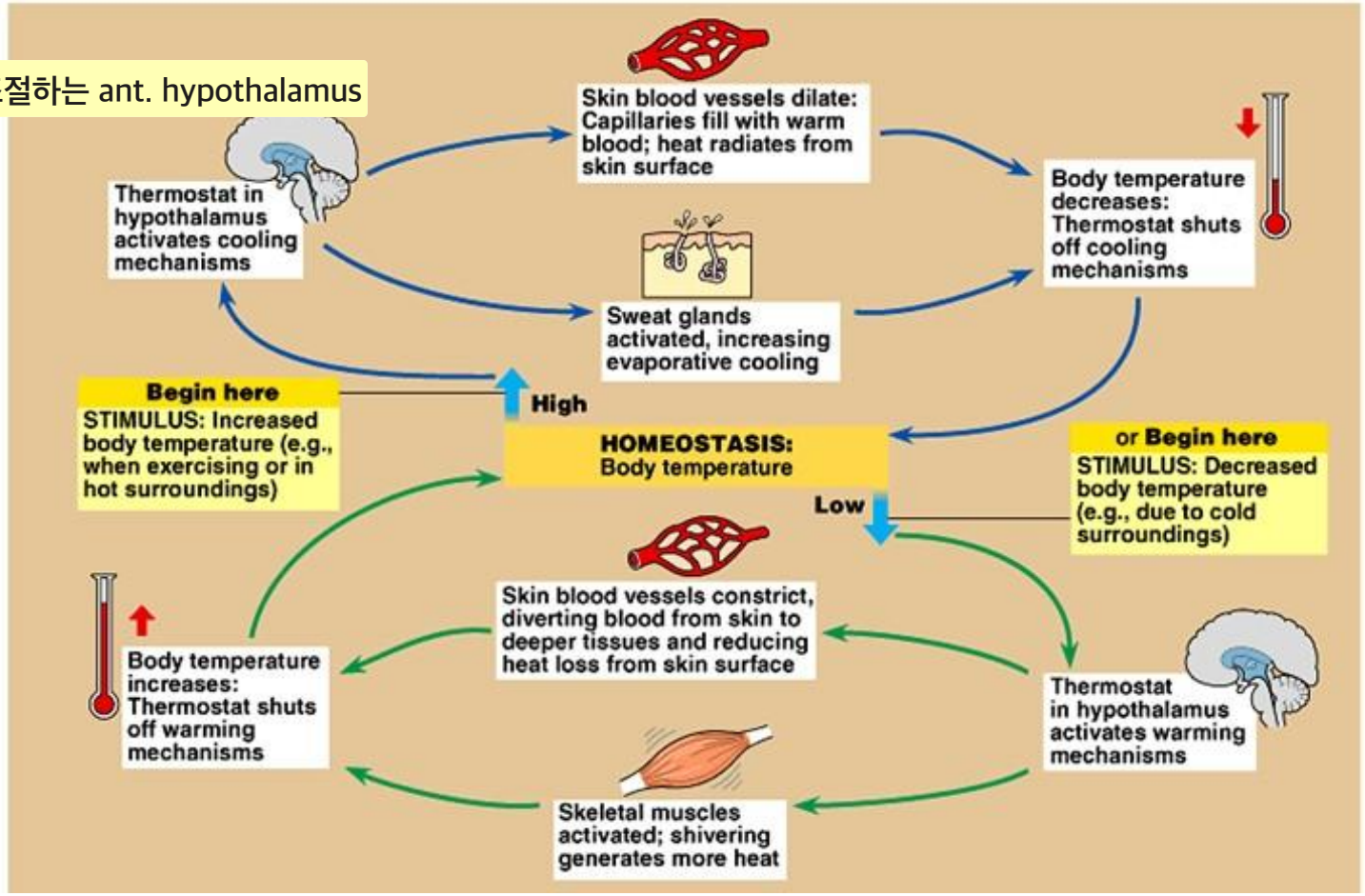
IV. Drowning

V. CO poisoning

VI. Snakebite

Temperature homeostasis (Thermoregulation)

체온을 조절하는 ant. hypothalamus



Thermoreceptor

- **Central thermoreceptors**
  - Hypothalamus, spinal cord, abdominal viscera, great veins
  - Cold < Warm receptors (1 : 50) → prevents hyperthermia

심부온도를 일정하게 유지.

- **Peripheral thermoreceptors**
  - Skin and mucosal surfaces
  - Cold > Warm receptors (10 : 1) → prevents hypothermia

내/외부 상황에서 온도를 감지

Body temperature

- **Core temperature** 특히 한랭질환에서...
  - Deep vital organs (brain, heart, liver, kidneys)
  - Measured at distal esophagus (most accurate), rectum, or bladder

- **Shell (skin) temperature** 피부 고막의 온도측정 · 내부 온도 (민감하게 변화)
  - Regulated by skin blood flow & ambient temperature
  - Acts as a thermal buffer to protect core temperature

Thermoregulation

감지

Stimulus
Core temp ↑ / ↓

→

Sensors
Skin + core thermoreceptors

→

Integrator
Preoptic anterior hypothalamus

→

열발산/보존

Effectors
Heat loss ↔ heat gain

Heat · core temp ↑

Skin vessels	Vasodilation (arterioles, AV anastomoses)
Sweat glands	Sweating → evaporation
Heart	HR ↑, cardiac output ↑
Splanchnic & renal vessels	Vasoconstriction → blood shunted to skin
Behavior	Seek shade, remove clothing, activity ↓
Endocrine (chronic)	Aldosterone↑, ADH↑ → Na^+/water retention

Heat · core temp ↓

반복적인 내려놈

Skin vessels	Vasoconstriction, cold-induced vasodilation
Skeletal muscle	Shivering
Endocrine	Catecholamines (acute), thyroid hormone (chronic)
Brown adipose tissue	Nonshivering thermogenesis (mainly infants)
Behavior	Seek shelters, add clothing, activity↑
Arrector pili	Piloerection (minimal in humans)

Exchange of energy with the environment

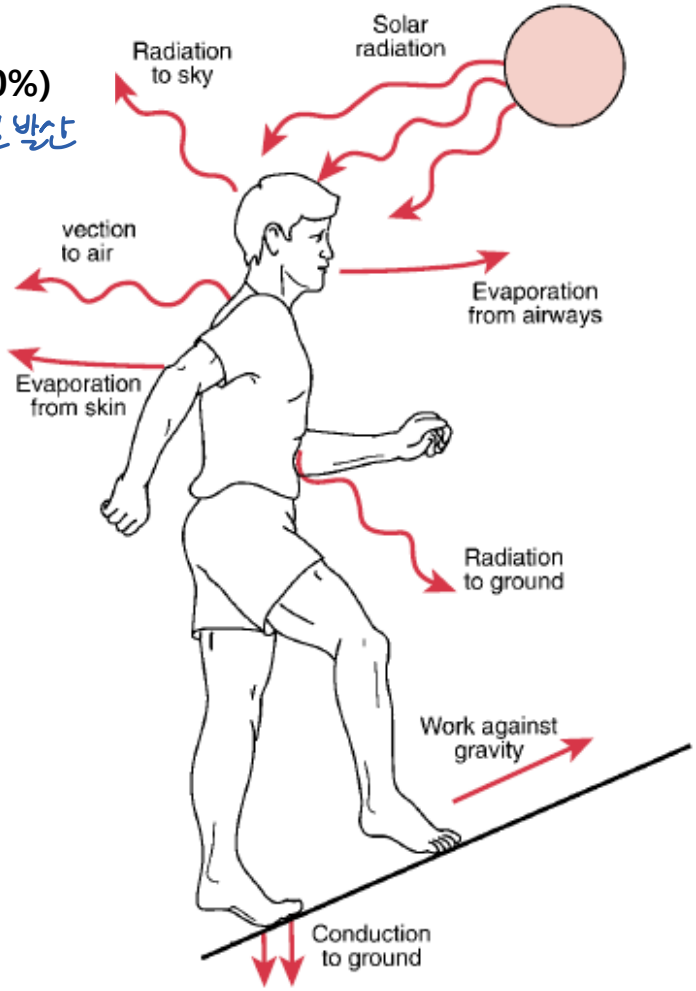
→ 일정한 체온을 유지

1. Radiation (~60%)

· 접촉 x 피부로 열 발산

3. Convection

· 공기나 흐름에 의해



2. Evaporation (~25%)

· 땀이 마르면서 열이 소실

4. Conduction

· 전도.

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Cold-related illness

- **Local cold injuries**
  - Non-freezing (wet, > 0˚C): Trench foot, Chilblains
  - Freezing (< 0˚C): Frostnip → Frostbite
  - Others: Cold panniculitis, Cold urticaria

- **Systemic cold injury**
  - Accidental hypothermia

Local cold injuries

•Non-freezing cold injuries

영상에서 발생

- Wet, cool conditions above freezing (0–15˚C): Trench foot, Chilblains

동창

- Treatment: Slow passive rewarming at room temperature

rewarm이면 충분

↘ inflammation 기전

• Freezing cold injuries

- Subfreezing temperatures (< 0˚C): Frostbite (동상)
- Treatment: Rapid active rewarming in 37–39˚C water bath



KNUH 경북대학교병원

Nonfreezing cold injuries
1. Trench foot (Immersion foot)

- Pathophysiology
  - Multifactorial
  - Prolonged exposure to **wet**, cold 습기·바람에서기등
  - m/c – military personnel have worn boots continuously 특히군인에서 많이 걸림

- Clinical feature

	1st stage (during cold exposure)	2nd stage (following cold exposure)	3rd stage (hyperemia)- rewarming	4th stage (following hyperemia)
Blood vessel	vasoconstriction	Peripheral blood flow ↑	Peripheral blood flow ↑ ↑ ↑	
Symptom	Numbness (m/c) No pain/swelling		Severe burning pain	
extremity	Bright red → pale No pain/swelling	Mottled pale blue	Hot, red Edema Necrosis, bullae (severe)	Normally recovery Gangrene (severe) 심하면
Duration		A few hrs (~ several days)	Wks ~ months	Wks ~ yrs

그렇지만 대부분은 잘 낫는다 (benign)









Nonfreezing cold injuries

1. Trench foot (Immersion foot)

- **Management**
  - Rewarm slowly at room temperature
  - Keep dry & elevated, cool fan in hyperemic stage
  - Neuropathic pain: Amitriptyline, Gabapentin
  - Tetanus prophylaxis (다리 open wound 가 있는 경우)

- **Prevention**
  - Well-fitting boots & change wet socks daily 예방 어려움

Nonfreezing cold injuries
2. Chilblains (Pernio)

- Pathophysiology: Localized inflammatory vasculopathy
- Epidemiology: More common in young women, children, outdoor workers, soldiers
- Common sites: Fingers, toes (m/c), ears, nose, cheeks 노부에 노출된 경우
- Risk factors: Raynaud phenomenon, SLE
- Clinical features: Pruritus & burning pain → violaceous plaques, edema 가려움


끝발톱의 edema 발생







Nonfreezing cold injuries
2. Chilblains (Pernio) - Management

- **General Principles**
  - Passive rewarming in room air, elevate & loose dressings
  - NO rubbing, massage, or direct heat; smoking cessation

- **Medical Treatment**
  - Topical corticosteroids (pruritus/erythema) & NSAIDs (pain)
  - Nifedipine (for severe or recurrent cases)
  - Tetanus prophylaxis if skin breakdown

Freezing cold injuries
Frostbite

심한경우 증상

- Freezing tissue injury: 4 phases (Prefreeze → Freeze-thaw → Stasis → Ischemia)

혈전

- **Epidemiology**
  - High-risk: Military, mountaineers, outdoor workers, homeless, elderly
  - Common sites: Ears, nose, cheeks, fingers, toes

특히 많이 발생

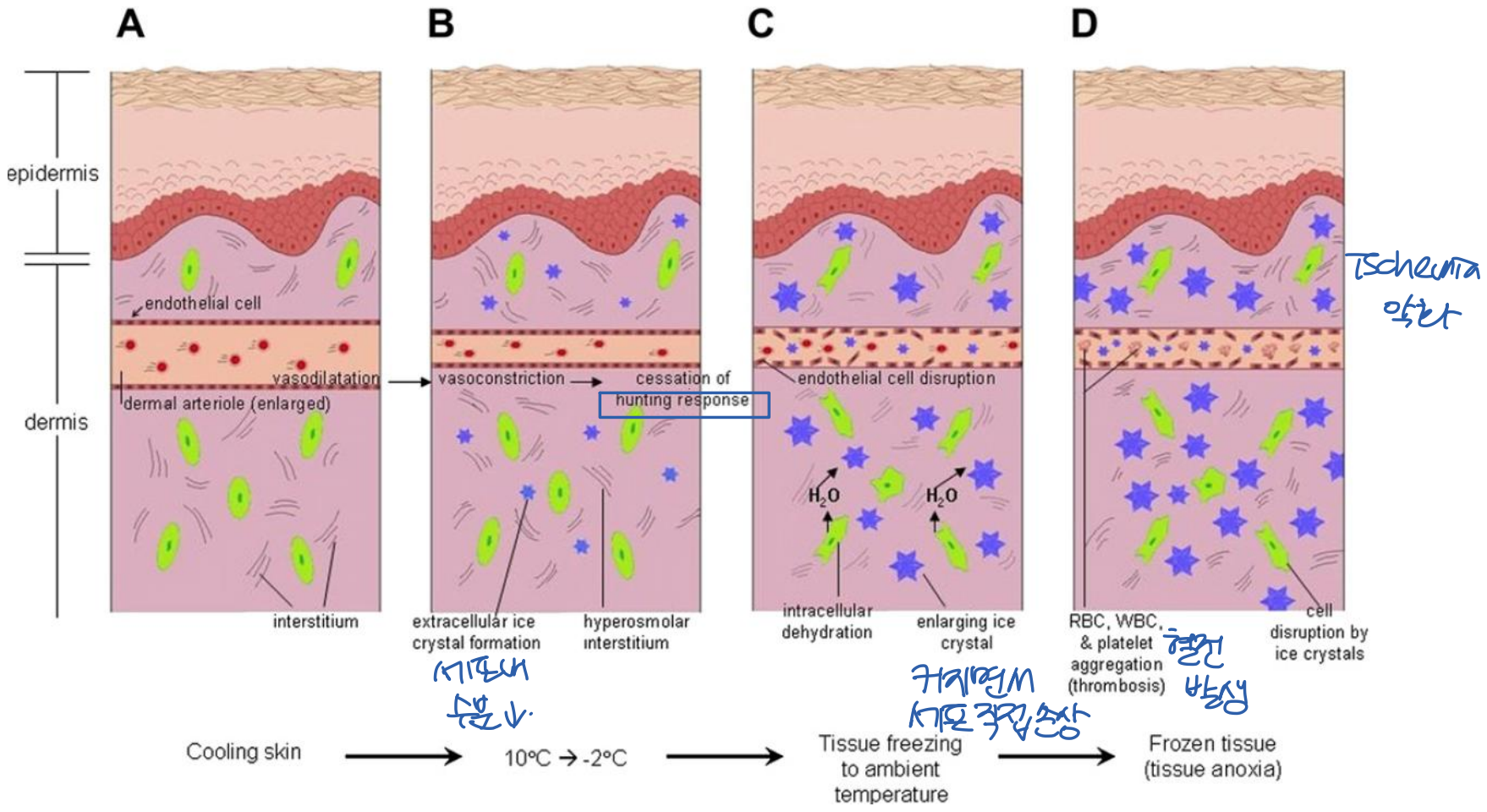
- **Risk Factors**
  - Windchill, high altitude, wet/constrictive clothing
  - Alcohol/smoking, Raynaud's, peripheral vascular disease, diabetes

병력 확인

Frostbite -
Pathophysiology

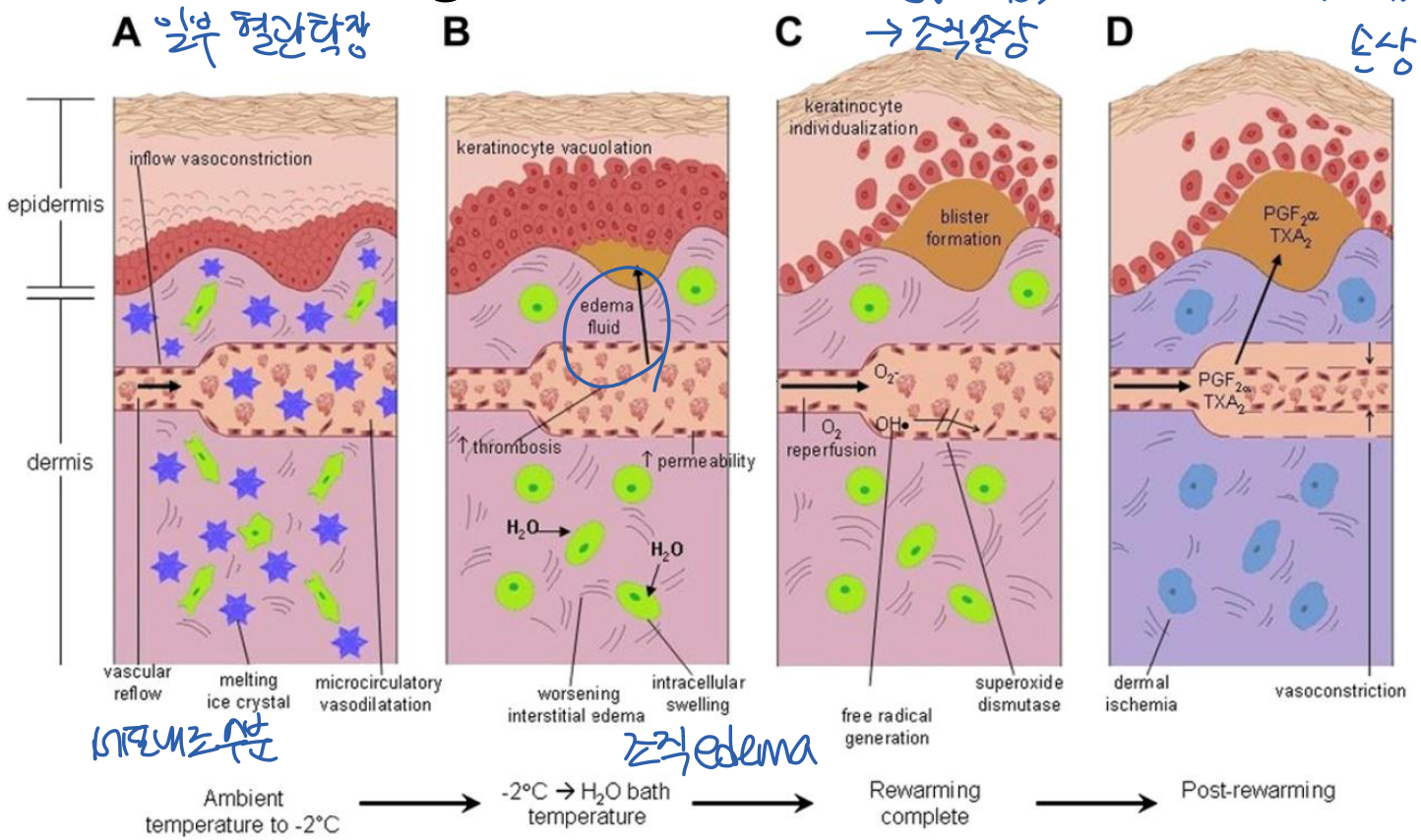
Direct cellular injury (ice crystals) + Indirect microvascular thrombosis

1. Tissue freezing 동결



Frostbite -
Pathophysiology

2. Tissue thawing 해동



Arachidonic acid cascade → vasoconstriction, platelet aggregation
→ venule and arterial thrombosis → ischemia, necrosis, dry gangrene

Frostbite
Clinical feature

- Frostnip: Reversible vasoconstriction (no deep ice crystals or tissue loss)

- **Clinical 4-Degree classification** (assessed **after** rewarming) 일반적으로 4단계로 나뉘어 있으나 구분이 힘듦

	superficial		Deep	
	1st degree	**2nd degree**	**3rd degree**	**4th degree**
Depth	Partial-thickness epidermis	Full-thickness epidermis ± Superficial dermis	Full-thickness skin + subQ tissue 혈관	Muscle, tendon, bone 까지.
Findings	• Numbness, erythema • Edema • No blister	물집 • **Clear/milky blisters** • Surrounding erythema, edema	• **Hemorrhagic blisters** • Blue-gray skin	• Mottled, cyanotic • Insensate • No blister • Minimal edema
Outcome	**Desquamation** **No tissue loss**	Eschar → heals **Usually no tissue loss**	**Tissue loss likely**	**Dry gangrene** **Amputation**

Tissue Injury



1st degree

· erythma



2nd degree

· blister



3rd degree

- hemorrhage blister
- blue - gray skin



4th degree

- necrosis

Frostbite

Prehospital Management 병원 전 단계

- Treat moderate/severe hypothermia & trauma first; hydrate
- Remove jewelry & tight gear
- Do **NOT** rub or walk on frozen foot 손상부위를 걷거나 만지기 x
- **Keep frozen if any risk of refreezing exists** 녹일수 없으면 그대로
- Field water rewarming (37–39°C) only if > 2h from care & no refreeze risk
  RISK 없는 경우
- Start oral Ibuprofen & opiate analgesia
  NSAIDs
- No routine O2 unless hypoxic (SpO2 < 88%) or altitude > 4,000 m
  불필요한 산소 x

Frostbite
ED Management

- **Rapid Water-Bath Rewarming**
  - **37–39°C circulating water (~30 min) until soft/pliable & red-purple**
  - No dry heat

- **Pain control**
  - IV opioids
  - Oral Ibuprofen

- **Blister & Wound Care**
  - Selectively drain clear tense blisters, leave hemorrhagic blisters intact
  - Topical Aloe vera q6h
  - Loose bulky dressings, elevation
  - Tetanus immunization

Frostbite

Perfusion Deficit Therapy & Disposition

- **Reperfusion Therapy for Deep Frostbite**
  - IV or IA tPA (< 24h from thawing, best < 12h; + Heparin/LMWH)
  - IV Iloprost (< 72h from thawing, Grades 2–4)

- **Imaging & Surgical Timing**
  - Viability imaging: Doppler, Angiography, 99mTc Bone Scan, or SPECT/CT
  - **Defer amputation** 1–3 months for demarcation (unless sepsis/wet gangrene)
  - Hospitalize all deep frostbite
  - Urgent fasciotomy if compartment syndrome

Hypothermia

systemic cold injury

증상과 matching

- **Definition: Core temperature < 35.0˚C** (Revised Swiss System)
  - Mild (Stage I, 32–35˚C): Alert, vigorous shivering
  - Moderate (Stage II, 28–32˚C): Verbal, shivering decreased/absent
  - Severe (Stage III, < 28˚C): Painful/Unconscious, vital signs present
  - Cardiac arrest (Stage IV, < 24–28˚C): Unconscious, no V/S

- **Accidental (unintentional)** vs. Therapeutic hypothermia (TTM)

상황지시 targeted

Etiology

- **Primary:** Overwhelming environmental cold exposure
- **Secondary:** Impaired thermoregulation 여러 원인들
  - Endocrine: Hypothyroidism, hypoadrenalism, hypopituitarism, hypoglycemia
  - Neurologic: Spinal cord transection, head trauma, stroke
  - Drugs: Ethanol, sedative-hypnotics, opioids, phenothiazines
  - Sepsis, severe burns, exfoliative dermatitis
  - Iatrogenic: aggressive fluid resuscitation

Clinical manifestation

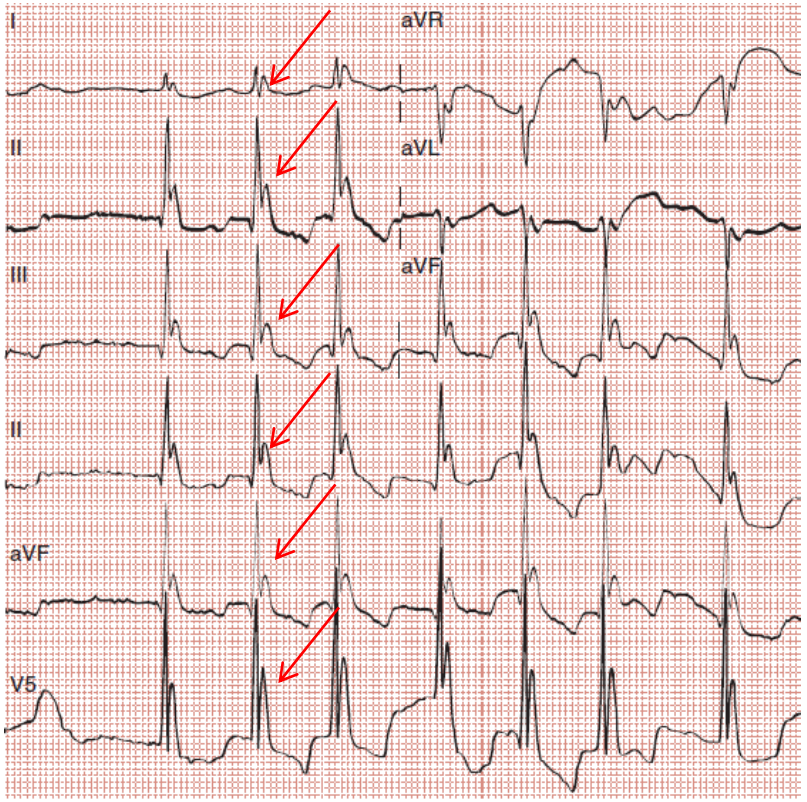
	Mild	Moderate	Severe
Neurologic	Confusion, slurred speech, impaired judgement, amnesia	Lethargy, hallucinations Loss of pupillary reflexes, EEG abnormalities	Loss of cerebrovascular regulation Decline in EEG activity Coma
Cardiovascular	Tachycardia CO ↑ SVR ↑	Progressive bradycardia CO/BP ↓ Atrial and ventricular **arrhythmias** **Osborn wave on ECG** (선택적)	Severe hypotension **Ventricular fibrillation (<28°C)** Asystole (<20°C) 저절로 or 약한 충격
Respiratory	Tachypnea Bronchorrhea	Hypoventilation O2/CO2 exchange ↓ loss of cough reflex	Pulmonary edema, apnea
Renal	**Cold diuresis** 소변을 보고 싶음 → dehydration 주의	**Cold diuresis** → IV fluid 필요	Decreased renal perfusion Oliguria
Musculoskeletal	Shivering (max) → 나두기	**Shivering ↓** 굳어진 체온 감소 muscle rigidity	**Patient may appear dead ("pseudo-rigor mortis")** 사후 경직

- Shivering stops below ~30 °C

Osborn(J) wave

- Positive deflection at J-point (< 32˚C)
- Suggestive, not pathognomonic
- DDx. 감별 필요
  - Benign early repolarization
  - Hypercalcemia
  - Brugada syndrome

그냥 이렇게 보일 수 있다~
중요한 소견은 아님



Core temperature measurement

- **Requires low-reading thermistor probe**
  - Esophageal probe (lower 1/3): Gold standard in intubated patients
  - Bladder / Rectal probe: Alternative in non-intubated patients (time lag)
  - Oral & standard infrared tympanic: Unreliable in hypothermia

Hypothermia Treatment
General management

- Basic or advanced life support
- Prevention of further heat loss 예방
- Rewarming ASAP, avoid refreezing
- Transport to an appropriate facility
- Treat underlying causes

· prosbite 동반↑

· ECMO 필요

Rewarming technique

- **Passive External (0.5–2˚C/h; Mild Stage I):** Dry clothes, blankets
  건조한 옷, 담요
- **Active External (1–2.5˚C/h; Stage II–IV):** Forced-air warming to **trunk**
  외부에서 공기주입
  - Caution: Warming extremities first causes afterdrop & VF
    ↓ trunk부위
- **Active Internal Rewarming (Severe Stage II–IV)**
  - Warmed IV fluids (40–42˚C) & warmed humidified O2 – (Stage II–IV)
  - Warm saline lavage (Thoracic, peritoneal, bladder) – (Stage III-IV)
  - ECLS (4–10˚C/h): VA-ECMO (1st choice) or CPB arrest/severe
    시행가능

Hypothermia Treatment (by staging)

- **Mild Hypothermia — Stage I (32–35˚C)**
  - Alert (GCS 15), vigorous shivering
  - Remove wet clothes + Passive external rewarming (blankets) 충분
  - Warm sweet drinks (if alert & swallowing intact)
  - Active movement allowed

Hypothermia Treatment (by staging)

core temperature

- **Moderate Hypothermia — Stage II (28–32˚C)**
  - Impaired consciousness (Verbal), shivering decreased/absent
  - Gentle horizontal handling → prevents 'Rescue Collapse' (VF)
  - Active external rewarming (to trunk) + Warmed IV fluids (40–42˚C)
  - Avoid early vasopressors (treat hypovolemia from cold diuresis first)

Active internal.

Unstable

fluid가 더 중요

Hypothermia Treatment (by staging)

Vital O, 의식 X

- **Severe Hypothermia — Stage III (< 28˚C, Unconscious, V/S present)**
  - Stable: Active external + minimally invasive internal rewarming
  - Unstable (SBP < 90, arrhythmia): Transfer to ECLS center (VA-ECMO)

이송

- **Hypothermic Cardiac Arrest — Stage IV (No V/S after 60sec check)**

60s check
촉진/맥박 X

  - Temp < 30˚C: Max 3 defibrillations & withhold ACLS drugs (Epinephrine) (ERC)
  - Temp 30–35˚C: Double drug interval (q 6–10 min); standard at ≥ 35˚C (ERC)
  - Continue CPR until ≥ 32–35˚C ('Nobody is dead until warm and dead!')

계속 시행

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Introduction

- Spectrum of disorder
  - From heat edema, heat cramp, heat exhaustion(minor) to heat stroke(major)

연속적임

- Disruption in regulation of body temperature

- Occurrence
  - Climate of high temperature/humidity
  - Summer heat waves, etc

온도↑, 습한 지역

Pathophysiology

- **Heat Transfer & Physical Limits**
  - Evaporation is main cooling mechanism (fails if humidity > 75%)
  - Radiation & convection fail when ambient temp > 35˚C

- **Physiologic Response to Heat Stress**
  - Cutaneous vasodilation, sweating, behavioral cooling
  - Elderly: higher sweating threshold & limited cardiac reserve

Pathophysiology

- **Effects of Medications** 중간
  - Anticholinergics, phenothiazines → inhibit sweating
  - Diuretics, alcohol → volume depletion
  - β-blockers, Ca-channel blockers → cardiovascular response ↓, sweat ↓
  - Cocaine, amphetamines, LSD, PCP → muscle activity ↑ / hypermetabolic state

Pathophysiology

- **Risk Factors**
  - Elderly, infants, obesity, poor fitness, lack of A/C, prior heat stroke

- **Heat Acclimatization** (takes 7–several wks, lost in 1–2 weeks)
  - Earlier & doubled sweating with Na+ conservation (Aldosterone ↑)
  - Increased plasma volume, lower HR, higher exercise tolerance

Pathophysiology

- **Classic (Non-exertional) Heat Injury** 더운곳에 노래있는 경우. ·발한↓
  - Elderly / chronically ill during heat waves

- **Exertional Heat Injury** ·발생↑
  - Young athletes, soldiers, outdoor workers

- **Confinement Hyperpyrexia** 더운곳에~.
  - Enclosed vehicles reach 54–60˚C in < 10 min

Fever vs. **Hyperthermia**

- Fever : **Elevated** hypothalamic set point (e.g., infection, cytokines)
- Hyperthermia : **Normal** set point → imbalance between heat creation and dissipation heat stroke

Heat edema

- Swelling of hands, feet, ankles in heat exposure

- Mechanism
  - Cutaneous vasodilation
  - Orthostatic pooling of interstitial fluid

- DDx.
  - CHF, DVT

- **Self-limited** 특별치료x.

- Treatment
  - Elevation of extremities 가능.
  - Compressive stockings
  - Diuretics: not effective

Heat cramp

자다가쥐난경우

- Painful muscle spasms (calves, thighs) during/after exertion

- Mechanism
  - Sweat Na+ loss + plain water intake → Dilutional hyponatremia

- Treatment & Prevention

water w. Na+

  - Cool rest + Oral 0.1–0.2% saline (sports drink) or IV 0.9% NS
  - Undiluted salt tablets: NOT recommended

Heat syncope

- Mechanism
  - Volume depletion
  - Peripheral vasodilation
  - Vasomotor tone ↓
- Elderly, poorly acclimatized individuals
- Evaluation for injury and causes of syncope (ACS, arrhythmia..)
- Treatment
  - Rest
  - Oral or intravenous rehydration

Heat exhaustion

열탈진

- Water & sodium depletion (precursor to heat stroke)

열사병의 전단계

- Clinical Manifestations
  - Fatigue, dizziness, headache, nausea/vomiting, profuse sweating
  - Tachycardia, orthostatic hypotension
- Key Distinction from Heat Stroke

감별

① - Core temp usually < 40˚C

② - **No sign of CNS dysfunction** ☆ 핵심

Heat exhaustion
Treatment

- **Treatment**
  - Rest in cool/shaded area, remove excess clothing
  - Oral electrolyte solution or IV isotonic crystalloids

의식O

- **Disposition**

대부분 회전 okay

  - Discharge after clinical recovery (2–3h observation)
  - Admit if elderly, comorbidities, or persistent symptoms

Heat stroke

- **Life-threatening emergency**

- **Diagnostic Hallmarks**

  atleast 1 → 심해지면

  - **CNS dysfunction** (Mandatory): Ataxia, delirium, seizures, coma

  - **Core temperature > 40˚C** (Not absolute if pre-cooled)

    대부분X . 오다가 cooled

- **Systemic Injury & Pitfalls**

  - Rhabdomyolysis, AKI, hepatic failure, DIC, ARDS

  - Anhidrosis is NOT required

Signs and symptoms of heat emergencies

	Heat cramps	Heat exhaustion	Heat stroke
Core temp	N ~ mild ↑	Usually <40℃	>40℃ (not absolute)
Sweating	Present	Profuse	Variable
CNS	Normal	**No** CNS dysfunction	**CNS dysfunction**
Other	Muscle cramps	Headache, malaise, N/V, orthostatic hypotension, tachycardia	Shock, rhabdo, AKI, liver injury, coagulopathy

Classic VS exertional heat stroke

어차피 치료가 같아서 구분의미X

	Classic	Exertional 하나도너…
Occurrence	Heat wave	sporadic
Age	Older	Young
General health	Predisposing illnesses/ medication use	Healthy 지병X
Lifestyle	Sedentary	Active
Sweating	Usually absent	Usually present 땀O
Skin	Hot, dry, reddish	Ashen, clammy
Coagulation status	Mild coagulopathy	DIC
ARF/Rhabdomyolysis/ Lactic acidosis	uncommon	common

Diagnosis

- Clinical diagnosis
  - **CNS dysfunction + heat exposure history** 신부활동
  - No diagnostic test for heat stroke 다른질환과 감별필요

- Core temperature: rectal or esophageal

- **Exclusion of other processes**

Differential diagnosis

모두 감별후 heat stroke

- Drug Toxicity: Anticholinergics, sympathomimetics, salicylates
- Drug Syndromes: Serotonin syndrome, NMS, Malignant hyperthermia, EtOH withdrawal
- Infections: Sepsis, meningitis/encephalitis, malaria
- Endocrine: Thyroid storm, DKA, pheochromocytoma
- Neurologic: Status epilepticus, cerebral hemorrhage

Heat stroke treatment

1st

- **Active Cooling > Passive Cooling**
  - Cooling rate ≥ 0.155˚C/min FAST
    → reach core temperature **< 39.0˚C within 30 min** of symptom recognition
    (afterdrop: core temperature may continue to fall after cooling steps)

- **Resuscitation & Organ Support**
  - A-B-C, cooled IV crystalloids 4°C saline 주입
  - Benzodiazepines for active seizures/shivering 선별 사용

- **Medications to avoid**
  - NO Dantrolene & NO Antipyretics (Acetaminophen, NSAIDs, Aspirin)
  - NO routine prophylactic anticonvulsants or antibiotics

malig. hypothermia

Cooling techniques

- **1st-Line Priority: Water Immersion** 목에 닿을 것
  - Ice-Water (1–5˚C: 0.178˚C/min) or Cold-Water (9–12˚C: 0.154˚C/min)
    그까지 방법만
- **Alternative Active Cooling (if immersion unavailable)** 대안
  - Tarp-Assisted Cooling (TACO), Cold shower, Ice sheets
  - Evaporative + convective cooling (water mist + large fan)
- **Adjunct Modalities (Combine with primary method)**
  - Cooled IV saline, Ice packs to neck/axilla/groin
    그외 보조

Cooling modalities

How fast?

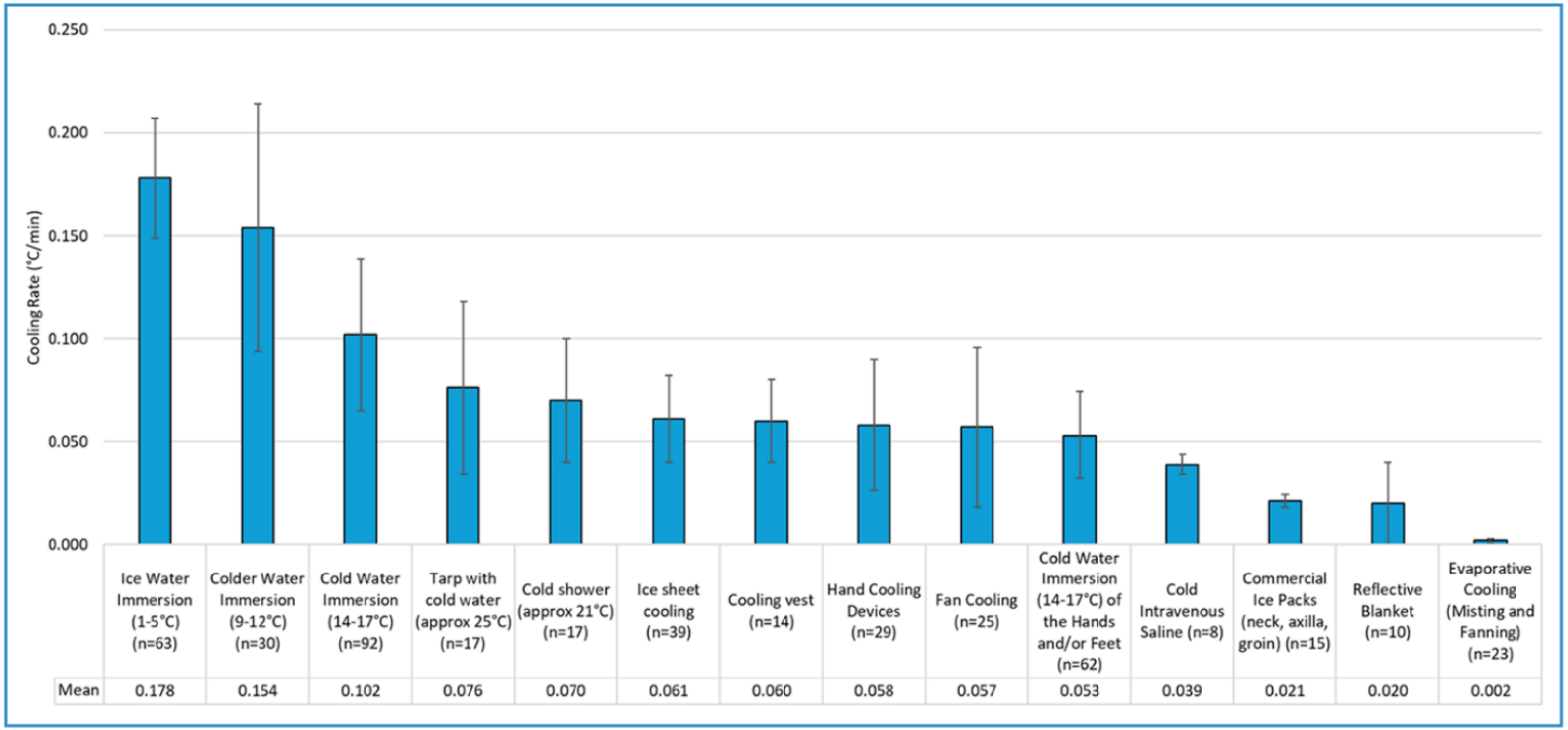


Figure 1. Mean weighted cooling rates by cooling method. To convert Celsius to Fahrenheit multiply by 1.8 and add 32.

Cooling modalities & 30min target

Cooling Method	Initial Core Temperature (°C) 40.0	40.6	41.1	41.7	42.2	42.8	43.3
	Final Core Temperature (°C)						
Ice Water Immersion (1-5°C)	34.7	35.3	35.8	36.4	36.9	37.5	38.0
Colder Water Immersion (9-12°C)	35.4	36.0	36.5	37.1	37.6	38.2	38.7
Cold Water Immersion (14-17°C)	36.9	37.5	38.0	38.6	39.1	39.7	40.2
Tarp with Cold Water (approx. 25°C)	37.7	38.3	38.8	39.4	39.9	40.5	41.0
Cold Shower (approx. 21°C)	37.9	38.5	39.0	39.6	40.1	40.7	41.2
Ice Sheet Cooling	38.2	38.8	39.3	39.9	40.4	41.0	41.5
Cooling Vest	38.2	38.8	39.3	39.9	40.4	41.0	41.5
Hand Cooling Devices	38.3	38.9	39.4	40.0	40.5	41.1	41.6
Fan Cooling	38.3	38.9	39.4	40.0	40.5	41.1	41.6
Cold Water Immersion (14-17°C) of the Hands and/or Feet	38.4	39.0	39.5	40.1	40.6	41.2	41.7
Cold Intravenous Saline	38.8	39.4	39.9	40.5	41.0	41.6	42.1
Commercial Ice Packs (neck, axilla, groin)	39.4	40.0	40.5	41.1	41.6	42.2	42.7
Reflective Blanket	39.4	40.0	40.5	41.1	41.6	42.2	42.7
Evaporative Cooling (Misting and Fanning)	39.9	40.5	41.0	41.6	42.1	42.7	43.2

Only

병원이에서는

그러므로 사용

Figure 2. Ability to reach target final core temperatures (°C) after 30 min based on each cooling modality and starting temperature. *Green* reflects a final core temperature less than 39°C (preferred), *yellow* reflects a final core temperature between 39°C and 40°C (borderline), and *red* reflects a final core temperature greater than 40°C (inadequate). To convert Celsius to Fahrenheit multiply by 1.8

Complication of heat stroke — early vs. late

TABLE 210-4 Complications of Heat Stroke

	Early	Late
Vital signs	Hypotension	—
	Hypothermic overshoot	
	Hyperthermic rebound	
Muscular	Rhabdomyolysis	—
Neurologic	Delirium/coma	Cerebral edema
	Seizure	Encephalopathy
		Persistent neurologic deficit
Cardiac	Heart failure	Myocardial injury
Pulmonary	Pulmonary edema	Acute respiratory distress syndrome
Renal	Oliguria	Renal failure, rhabdomyolysis
GI	—	Intestinal ischemia or infarction
		Pancreatic injury
		Hepatic dysfunction
Metabolic	Hypokalemia	Hyperkalemia
	Hypernatremia	Hypocalcemia
	Hyponatremia	Hyperuricemia
Hematologic	—	Thrombocytopenia
		Disseminated intravascular coagulation

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목차

I. Temperature regulation

II. Cold-related illness

III. Heat-related illness

IV. Drowning

V. CO poisoning

VI. Snakebite

Drowning

- Definition: Primary respiratory impairment from submersion/immersion
  - Classified as Fatal vs. Non-fatal drowning (Do NOT use 'near/wet/dry drowning')

- Epidemiology
  - 236,000~370,000 deaths/yr worldwide (1st injury death < 15 yrs)
  - Peak ages: < 5 yrs (highest), 15–24 yrs (2nd), elderly (3rd)

Drowning
Pathophysiology

- **Outcome ∝ degree of hypoxic insult to the CNS**

- Diving reflex
  - Bradycardia, apnea, peripheral vasoconstriction, central shunting of blood flow
  - Transient protection during submersion
  - Strongest in infants < 6months, **adults not protective**
  - Better outcome of cold water drowning: rapid CNS cooling, not diving reflex

Drowning
Pathophysiology

- Surfactant washout → Alveolar collapse, V/Q mismatch, & ARDS
- Hypoxemia & metabolic acidosis → Multi-organ / CNS damage
- Water/gastric acid/contaminated water aspiration → chemical pneumonitis/bacterial pneumonia
- Electrolyte changes & hemolysis: rare in survivors, DIC may occur following severe hypoxic insult

Drowning
Prehospital care

보통 hypoxia가 원인 · spasm 등.

- Resuscitation Sequence
  - Trained rescuer / Healthcare provider: **A-B-C** (**2 rescue breaths first**, then 30:2 CPR)
  - Untrained / General lay rescuer: C-A-B

- Heimlich maneuver (abdominal thrusts): contraindicated

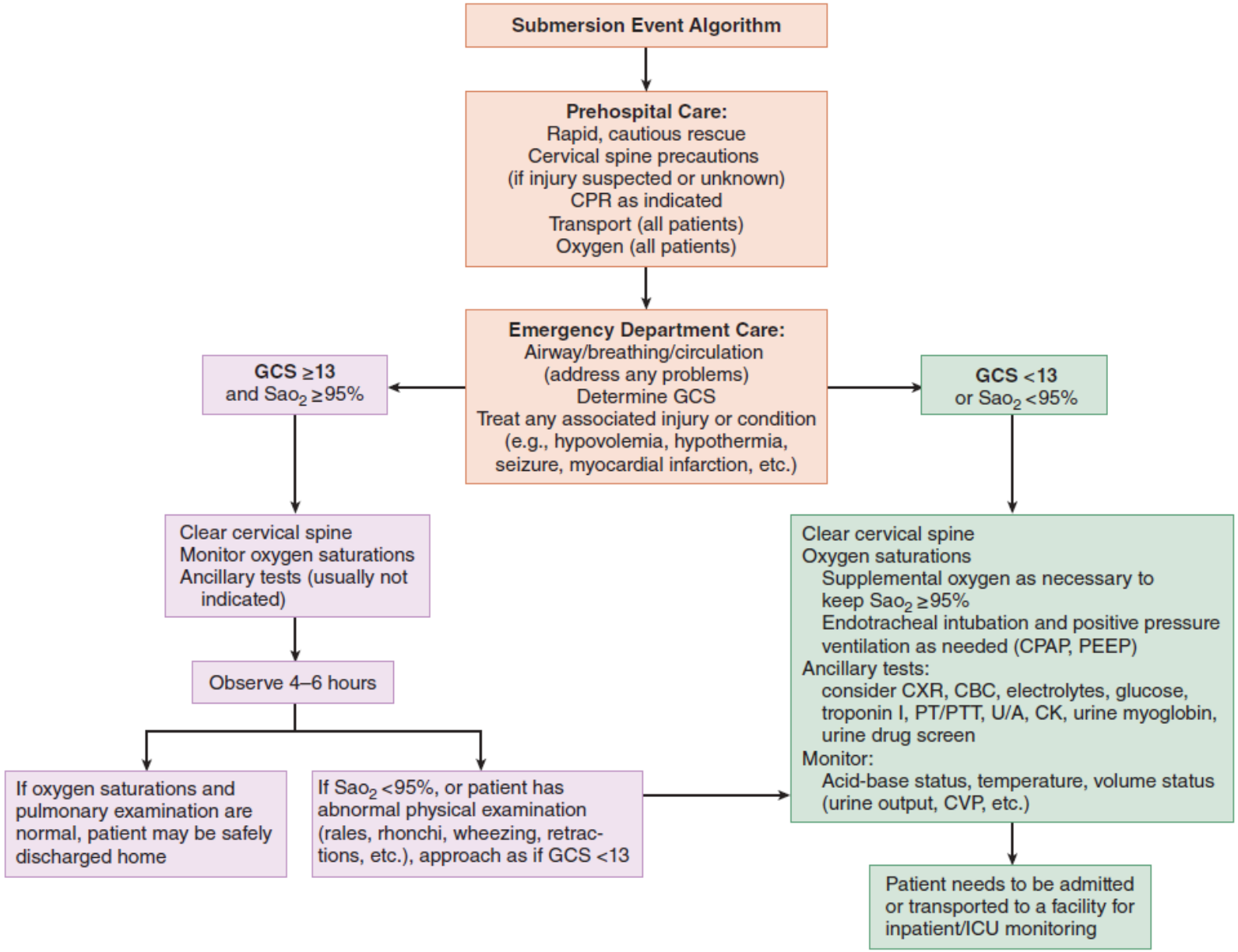
- Refer to ED if: Any LOC, apnea, rescue breathing, cyanosis, or dyspnea

Drowning
ED management

- Pulmonary Support
  - High-flow O2 (target SpO2 ≥ 94–95%)
  - Early CPAP/BiPAP or Intubation + PEEP

- Cervical Spine Immobilization: NOT Routine
  - Perform ONLY if diving, high-impact water sports, or trauma signs

- Associated Care & Prognosis
  - No routine prophylactic antibiotics or steroids
  - Observe mild cases 4–6 hours before discharge



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Carbon Monoxide(CO) poisoning

- Epidemiology
- US: 2nd most common fatal poisoning (after drug overdose), leading non-drug poisoning death
- Korea: 1st cause of toxic gas poisoning (accidental, & intentional)

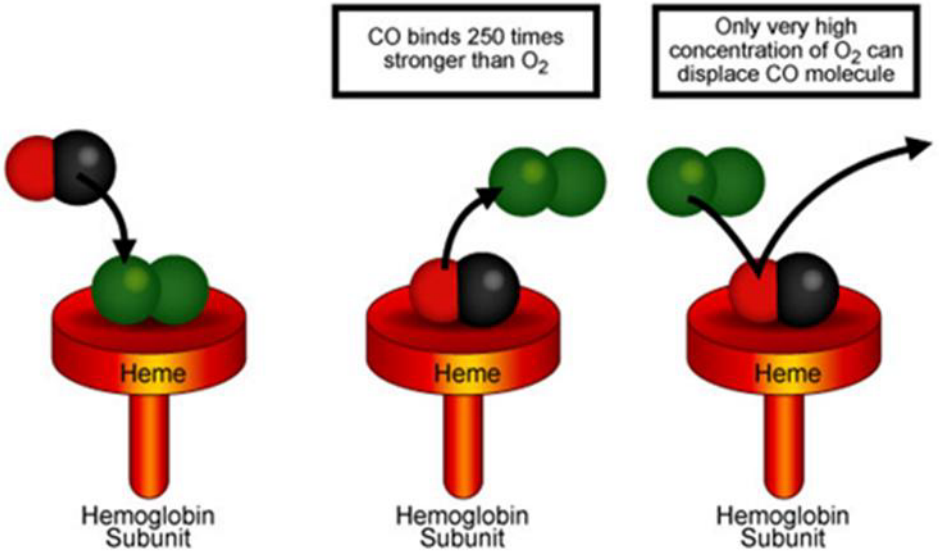
- Common sources of CO
  - Structure fires, faulty gas boilers/heaters, camping charcoal/propane stoves, coal briquettes
  - Vehicle/generator exhaust in enclosed spaces, methylene chloride (paint stripper → hepatic CO)

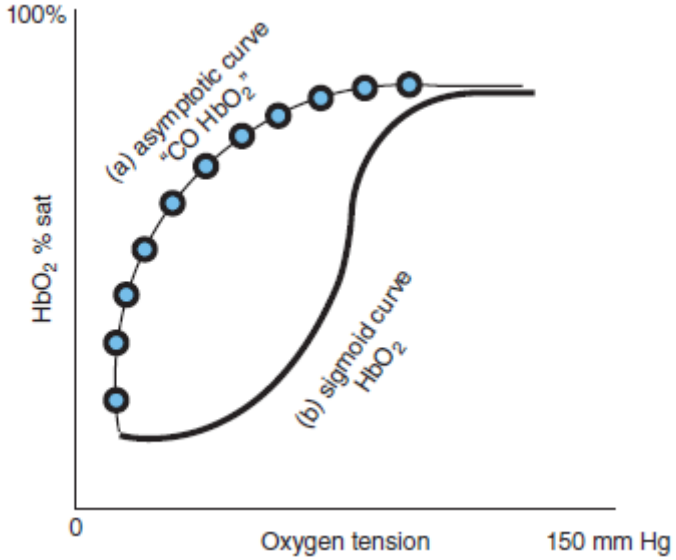
Carbon Monoxide(CO) poisoning
Characteristics of CO

- Odorless, colorless, nonirritating gas
- Normally present in air at 10 ppm or less. (Urban area > Rural area)
- Normal physiologic level of COHb
  - Healthy non-smoker: ~ 2% (typically less than 1 %)
  - Active smoker: 4 ~ 10%

Carbon Monoxide(CO) poisoning
Carboxy hemoglobin

- Hypoxic: Hb affinity 200–250× O2 → COHb + Leftward shift (Fetal HbF > HbA)
- Cellular: Inhibits Cytochrome c oxidase → ROS & lipid peroxidation (causes DNS)





Carbon Monoxide(CO) poisoning
Clinical feature

TABLE 222-2 Signs and Symptoms of Acute Carbon Monoxide Poisoning

Headache	Visual disturbances
Vomiting	Confusion
Ataxia	Dyspnea/tachypnea
Seizure	ECG changes/dysrhythmias
Syncope	Retinal hemorrhage
Chest pain	Bullous skin lesions
Focal neurologic deficit	

- History may be obviously suggestive

- Consider CO poisoning in the differential diagnosis of coma, mental status changes, and elevated anion gap metabolic acidosis or lactic acidosis

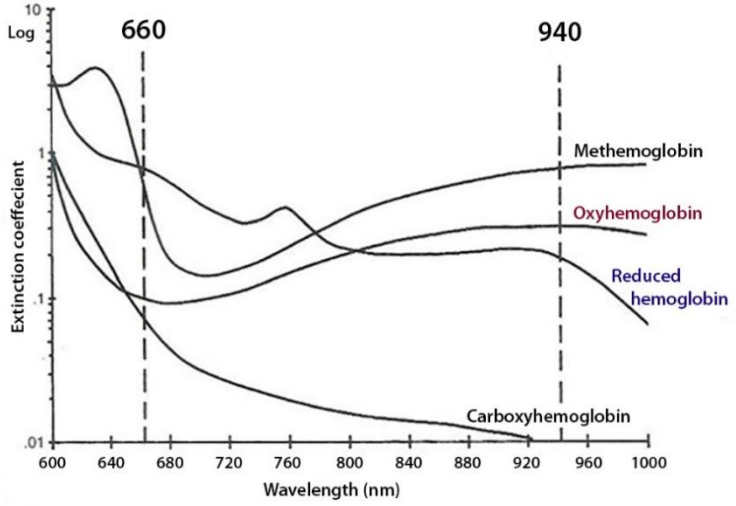
Carbon Monoxide(CO) poisoning
Clinical feature

- Cherry-red skin: Classic sign, but rare in living patients (< 2–3%, mostly post-mortem)



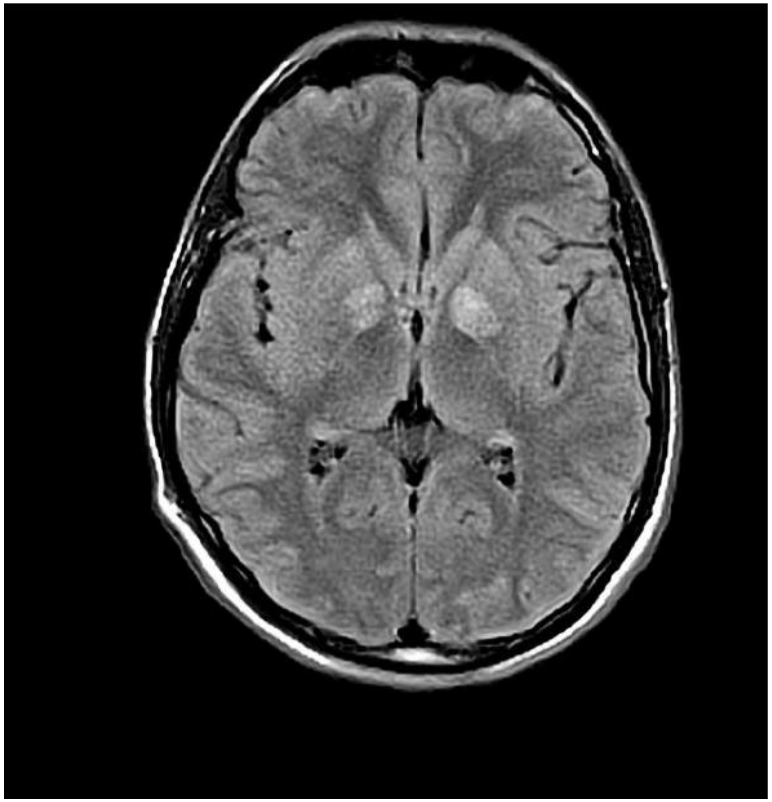
Carbon Monoxide(CO) poisoning
Diagnosis

- **Confirm with CO-oximetry (VBG / ABG) – COHb level**
  - Pulse oximetry (SpO2): falsely normal
  - Standard ABGA PaO2: normal (dissolved O2)
  - COHb level ≠ clinical severity
- **End-Organ Workup**
  - ECG, Troponin, Lactate, CPK, Brain MRI/CT



Carbon Monoxide(CO) poisoning

Radiologic image





Carbon Monoxide(CO) poisoning
Treatment

- **Start 100% Normobaric O2 (Non-rebreather mask) immediately!**
  - 100% O2 until asymptomatic & COHb < 3% (ED observation 4–6h)
- **Mild Symptoms (No high-risk features)**
- **High-Risk Features**
  - **Hyperbaric Oxygen Therapy** or high-flow normobaric O2 therapy

TABLE 222-3 Common Indications for Referral for Hyperbaric Oxygen Treatment

Loss of consciousness/syncope	Acute myocardial ischemia
Altered mental status/confusion	Cardiovascular dysfunction/dysrhythmia
Seizure	Hypotension
Coma	Severe metabolic acidosis
Focal neurologic deficit	Carboxyhemoglobin level ≥25%
Pregnancy carboxyhemoglobin level ≥15%	

Carbon Monoxide(CO) poisoning

Hyperbaric oxygen therapy (HBOT) 2-2.5 atm↑

반감기 단축 가능

- COHb Half-Life: Room air (4–5h) → 100% O2 (60–90 min) → HBOT (20–30 min)
- Prevents Delayed Neurologic Sequelae (DNS) by inhibiting lipid peroxidation





Carbon Monoxide(CO) poisoning
special situation

- **Delayed Neurologic Sequelae**
  - Occurs after 2–40 day 'lucid interval' in 15–40% of survivors
  - Cognitive decline, personality change, Parkinsonism, gait ataxia, incontinence
  - Risk Factors
    - Age > 36–50 yrs, exposure > 6h, LOC / GCS < 9, seizure, leukocytosis
  - Educate patient/family at discharge & schedule 2–4 wk follow-up

목차

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Snakebite

- **Venomous Snakes in Korea (국내 독사 4종)** 독사 여부만 확인
  - **Viperidae (살모사과)**
    - Gloydius brevicaudus (살모사), G. ussuriensis (쇠살모사), G. intermedius (까치살모사)
  - **Natricidae (유혈목이과)** → 차이
    - Rhabdophis tigrinus (유혈목이/꽃뱀)

사진X. 학명X



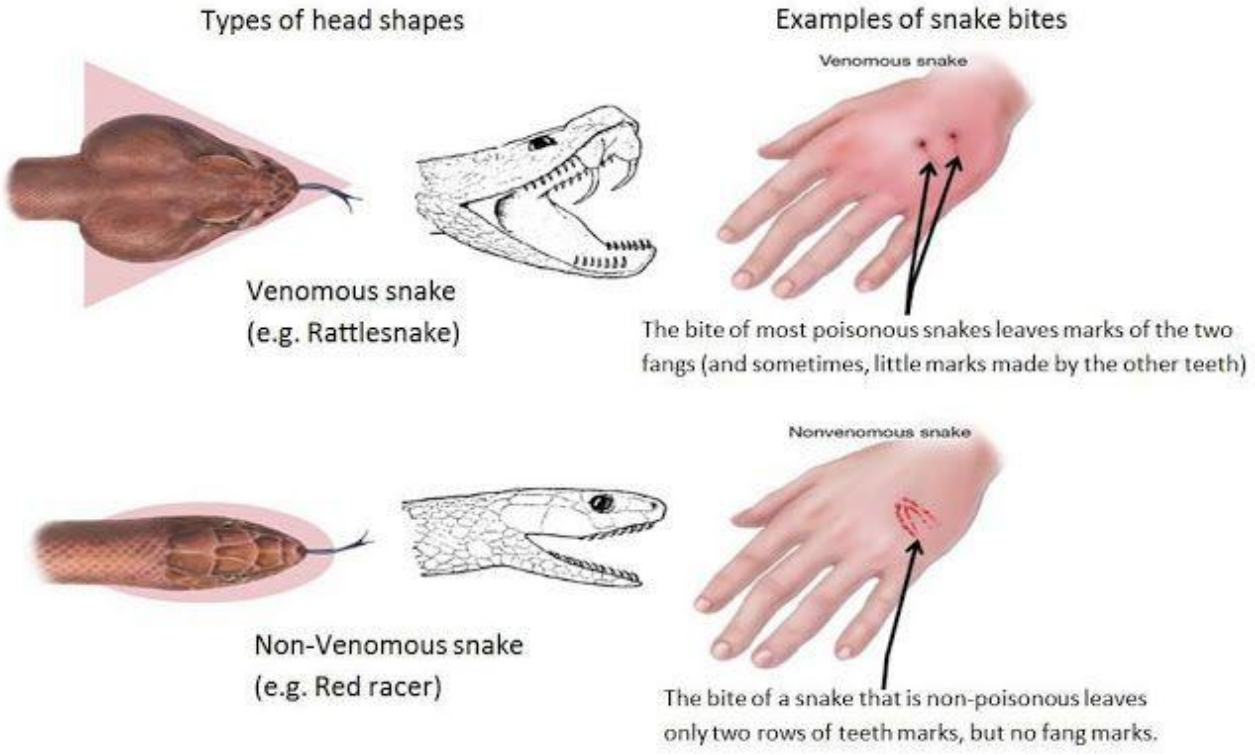



Snakebite
Clinical features

- Local tissue injury
  - pain, edema, ecchymosis, fang marks, hemorrhagic blebs

- Systemic symptoms
  - Nausea, vomiting, weakness (general), dizziness, muscular fasciculation
  - Gingival bleeding, hemolacria, GI bleeding, cerebral hemorrhage
  - Rhabdomyolysis, acute renal failure (간중 organoi damage)
  - Dyspnea, tachypnea, tachycardia, hypotension, decreased mentality, shock

Snakebite — Venomous vs. Non-venomous Snake



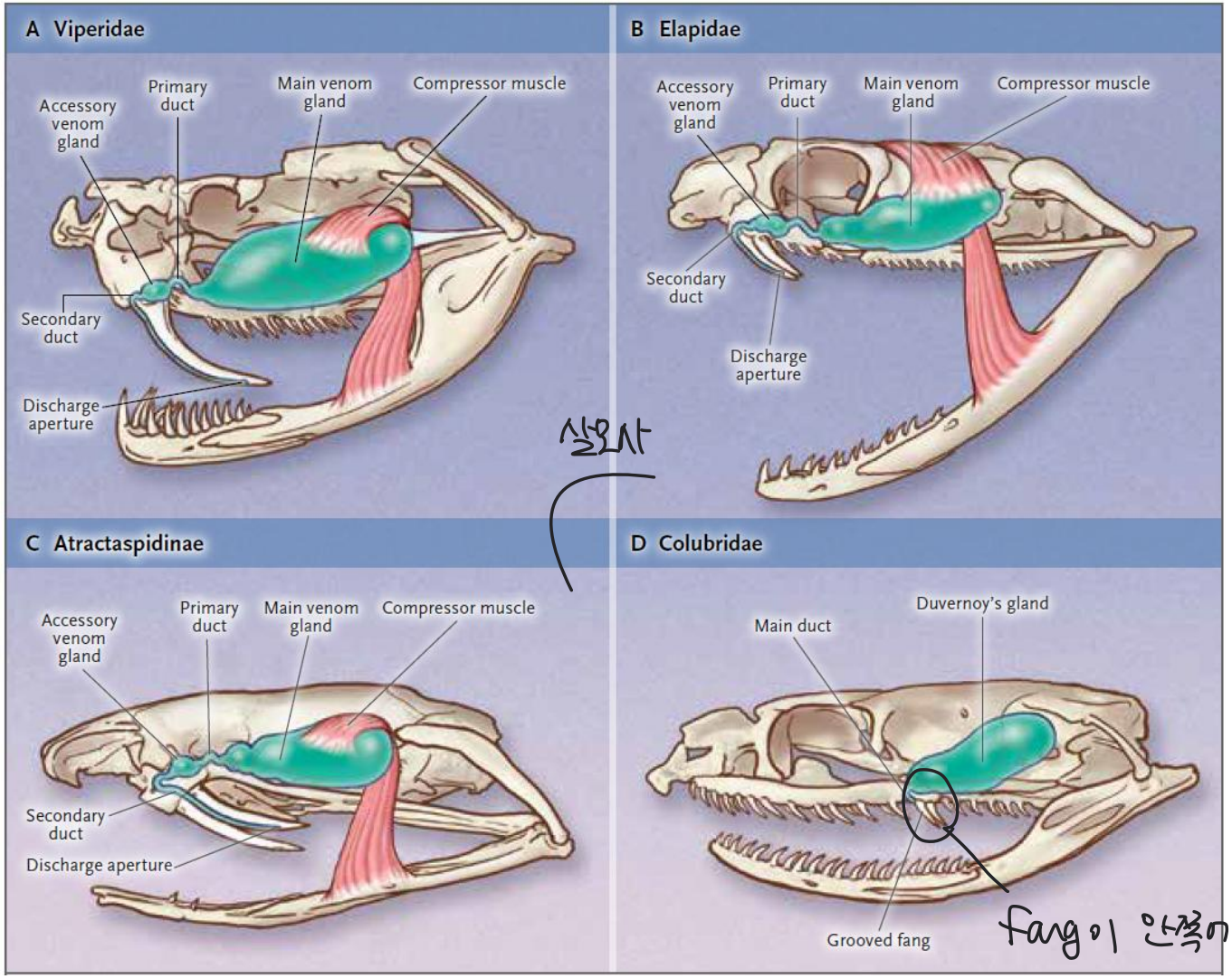
fang marks



- **Do NOT** exclude envenomation by head shape or fang marks alone!
  - Gloydius may leave 1 mark
  - Rhabdophis tigrinus has a round head & rear fangs.

Snakebite

Venom delivery systems of snakes



Snakebite
Clinical features

Fang marker의 추가 중요



Snakebite
Clinical features

관찰필요 + 다른위치~



Snakebite
Treatment – First Aid 현장처치

- **Recommended First Aid**
  - Calm patient & prohibit walking (No ambulation)
  - Remove rings, watches, tight shoes before swelling starts
  - Splint in neutral position at/below heart level & transport to ED

- **Strictly Contraindicated**
  - NO incision, suction, arterial tourniquet, ice packs, or alcohol

Snakebite
Treatment – ED management

살모사과에만 효과.

- **Antivenom (코박스 건조살무사항독소주)**
  - Indications: Progressive swelling, coagulopathy, systemic/neuro signs
  - Timing & Route: < 3–4h optimal; IV (1st) or IO (2nd)
  - Initial Dose: Mild–Mod 1 vial; Severe 2 vials
  - Pediatric dose = Adult dose 용량 차이X
  - Ineffective for Rhabdophis tigrinus (유혈목이 → FFP/Cryo 치료)



Snakebite
Treatment – ED management

- **Supportive ED Care**
  - Immobilize limb & tetanus booster
  - No routine antibiotics, steroids, or NSAIDs



- **Compartment Syndrome in snakebite**
  - 1st-line: Antivenom (± elevation, mannitol)
  - Fasciotomy: ONLY if pressure > 30–40 mmHg despite antivenom & hemostasis

Snakebite
Disposition

- Observe all bites for ≥ 6–8 hours with repeat coagulation labs
- Discharge: Dry bite (no progression & normal labs after 6–8h)
- Admission
  - Mild–Moderate envenomation → General ward
  - Severe envenomation or Antivenom given → ICU
- Follow-up: Monitor for infection & Serum sickness (1–2 wks post-antivenom)